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National University of Computer and Emerging Sciences

Lahore Campus

Data Communication Networks (EE2007)	**Final Exam**
Date: December 23, 2024	
Course Instructor:	**Total Time (Hrs):** 3
Dr. Saima Zafar	**Total Marks:** 100
Do not write below this line	**Total Questions:** 5

CLO # 1:

Q1: A. Answer the following Question. **[marks: 5+5]**

Consider a point-to-point link 2 km in length. (i) At what bandwidth would propagation delay (at a speed of 2×10^8 m/sec) equal transmit delay for 100-byte packets? (ii) Repeat for 512-byte packets?

Q1: B. Answer the following Questions. **[marks: 5+5]**

1) Draw a neatly labeled figure of a Network Model for the Internet.
2) What is the purpose of a DNS system? Define the different types of DNS servers that form the system.

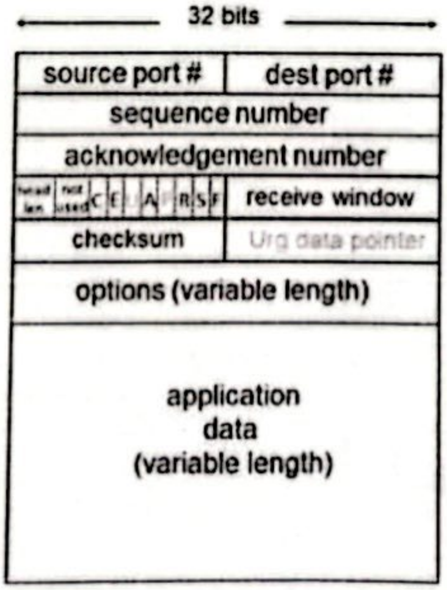
CLO # 2:

Q2: A. Answer the following Question. **[marks: 10]**

Suppose that the two measured SampleRTT values are 95 ms and 110 ms. Compute the EstimatedRTT after each of these SampleRTT values is obtained, using a value of $\alpha = 0.125$, and assuming that the value of EstimatedRTT was 90 ms just before the first of these two samples were obtained.
Also, compute the DevRTT after each sample is obtained, assuming a value of $\beta = 0.25$ and assuming the value of DevRTT was 4 ms just before the first of these two samples were obtained.
Finally, compute the TCP TimeoutInterval after each of these samples is obtained.

Q2: B. Answer the following Question. **[marks: 10]**

In the given TCP segment structures, explain the use of all of the fields indicated.

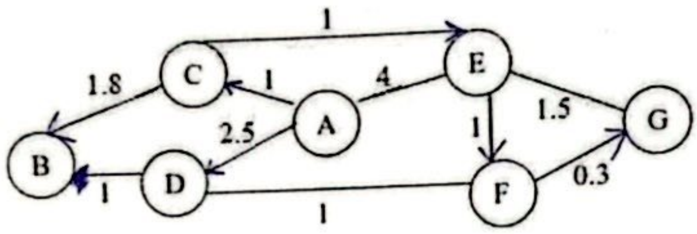


CLO # 3:

[marks: 10]

Q3: A. Answer the following Question.

For the given network, use Dijkstra's shortest path algorithm to compute the shortest path from A to all network nodes. You need to create the Routing Table for node A along with equations for each iteration.



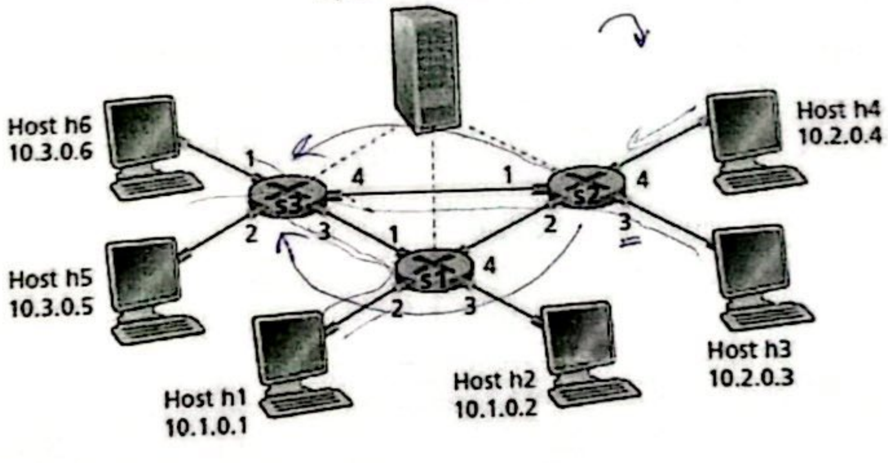
[marks: 10]

Q3: B. Answer the following Question.

Consider again the SDN OpenFlow network shown in Figure. Suppose that the desired forwarding behavior for datagrams arriving from hosts h3 or h4 at s2 is as follows:

- any datagrams arriving from host h3 and destined for h1, h2, h5 or h6 should be forwarded in a clockwise direction in the network;
- any datagrams arriving from host h4 and destined for h1, h2, h5 or h6 should be forwarded in a counter-clockwise direction in the network.

Specify the flow table entries in s2 that implement this forwarding behavior.



2, 1, 5, 6

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CLO # 4:

Q4: A. Answer the following Question. **[marks: 10]**

Suppose we want to transmit the message 11001001 and protect it from errors using the CRC polynomial $x^3 + 1$. Use polynomial long division to determine the message that should be transmitted.

1001
$x^3\ x^2\ x^1\ x^0$

Q4: B. Answer the following Questions briefly. **[marks: 10]**

We have come across three generations of Carrier Sense Multiple Access protocols -- the original CSMA, CSMA/CD, and CSMA/CA. Based on your knowledge of these variants of CSMA MAC protocols, address the following questions:

1. How does CSMA work in principle?
2. What is the problem in CSMA that CSMA/CD is trying to resolve?
3. What is the problem in CSMA/CD which CSMA/CA is trying to resolve?

CLO # 5:

Q5: A. Answer the following Questions briefly. **[marks: 10]**

1. In the Nyquist formula for maximum data rate in a noiseless channel, $max\ data\ rate = 2Blog_2V$, what does *V* represent in (a) analog signaling and (b) in digital signaling?
2. What are the two desirable properties for line coding schemes?
3. Define Baud Rate, Symbol Rate, and Bit Rate.
4. Define QPSK (Quadrature Phase Shift Keying).
5. Is CDMA a form of spread spectrum communication? Justify your answer.

Q5: B. Answer the following Questions. **[marks: 5+5]**

1) A CDMA receiver gets the following chips: (-1+1-3+1-1 -3 +1 +1). Assuming the chip sequences defined below, which stations transmitted, and which bits did each one send?

A = (−1 −1 −1 +1 +1 −1 +1 +1)
B = (−1 −1 +1 −1 +1 +1 +1 −1)
C = (−1 +1 −1 +1 +1 +1 −1 −1)
D = (−1 +1 −1 −1 −1 −1 +1 −1)

2) Generate the signals for the following line coding schemes: NRZ, NRZI, and Manchester, for the bit pattern: 1001011. Clock signal is as follows: